CS2323 Computer Architecture

Homework-1: RISC-V Assembly and Binary Numbers
MA24BTECH11022

Q1. Write an assembly instruction to achieve the given functionality, defined using C-language syntax (only 1 instruction to be used). Explain in brief. [6 marks]

(a) $x8 = x5 - 5$
addi x8, x5, -5
We need to add a constant to the value stored in x5, so we use addi instruction

(b) $x5 = x3 * 8$
slli x5, x3, 3
Multiplying by 8 is equivalent to shifting left by 3 bits, so we use the slli instruction, assuming we are dealing with signed numbers in the longer run

(c) $x19 += x10$
add x19, x19, x10
x19 += x10 is equivalent to x19 = x19+x10, hence we use the add instruction

(d) $++x15$
addi x15, x15, 1
++x15 is equivalent to x15 = x15+1 and noting that 1 is a constant, we use the addi instruction

(e) $x9 = x15/4$
srai x9, x15, 2
Dividing a value by 4 is equivalent to right shifting it 2 times. Noting that 4 is a constant and assuming that in the longer run we are dealing with signed integers, we use srai instruction

(f) $x12 = x19 + 24$
addi x12, x19, 24
We need to add a constant to the value stored in x19, so we use addi instruction.

Q2. Consider an array M consisting of 8 byte integers. The base address of M is stored in register x5. Write the assembly code that achieves each operation given below. [1+1+1+2+2 marks]

(a) $M[12] = M[20] + 100$
ld x1, 160(x5)
addi x1, x1, 100
sd x1, 96(x5)

(b) $M[20]++$
ld x1, 160(x5)
addi x1, x1, 1
sd x1, 160(x5)

(c) swap $M[5]$ with $M[12]$
ld x1, 40(x5)
ld x2, 96(x5)
sd x2, 40(x5)
sd x1, 96(x5)

(d) make the first 32-bits (from MSB side) of $M[4]$ as 0
ld x1, 32(x5)
slli x1, x1, 32
srli x1, x1, 32
sd x1, 32(x5)

(e) swap the most significant 32-bits of $M[2]$ with its least significant 32-bits
ld x1, 16(x5)
srli x2, x1, 32
slli x1, x1, 32
or x1, x1, x2
sd x1, 16(x5)

Q3. Write the following decimal numbers in their 2's complement representation, using 8-bits. Show your calculations. [4 marks]

(a) +23
$23 = 16 + 7 = 16 + 4 + 3 = 16 + 4 + 2 + 1 = 00010111$

(b) -1
Let $x = 1$. Then $\bar{x} + 1 = -x$.

$$\begin{aligned} x &= 00000001 \\ \Rightarrow \bar{x} &= 11111110 \\ \Rightarrow \bar{x} + 1 = -x &= -1 = 11111111 \end{aligned}$$

(c) +255
In 2's complement, we can only represent numbers from -128 to 127 using 8 bits. So we cannot write 255 using the given resources.

(d) -128
Let $x = 127$. Then $\bar{x} + 1 = -x$.

$$\begin{aligned} x &= 01111111 \\ \Rightarrow \bar{x} &= 10000000 \\ \Rightarrow \bar{x} + 1 &= 10000001 \\ &= -127 \end{aligned}$$

Now $-128 = -127 - 1 = 10000001 - 1 = 10000000$.

Q4. Write the equivalent decimal number for given numbers in 2's complement format. Show your calculations. [3 marks]

(a) 11010100
The first bit is 1, so the number is negative. Let the number be $-x$.

$$\begin{aligned} -x &= 11010100 \\ \Rightarrow \bar{x} &= 11010011 \\ \Rightarrow x &= 00101100 \\ &= 2^5 + 2^3 + 2^2 \\ &= 32 + 8 + 4 \\ &= 44 \end{aligned}$$

Therefore, the number is -44 in decimal.

(b) 00101011
The first bit is 0, so the number is non-negative.

$$\begin{aligned} x &= 00101011 \\ &= 2^5 + 2^3 + 2^1 + 2^0 \\ &= 32 + 8 + 2 + 1 \\ &= 43 \end{aligned}$$

Therefore, the number is 43 in decimal.

(c) 11111110
The first bit is 1, so the number is negative. Let the number be $-x$.

$$\begin{aligned} -x &= 11111110 \\ \Rightarrow \bar{x} &= 11111101 \\ \Rightarrow x &= 00000010 \\ &= 2^1 \\ &= 2 \end{aligned}$$

Therefore, the number is -2 in decimal.